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In [30]: # Import Libraries
import numpy as np
import pandas as pd
from sklearn.model_selection import train_test_split
from sklearn.linear_model import LinearRegression
from sklearn.metrics import mean_squared_error, r2_score

# Sample House Price Dataset
data = {
    "Area": [1000, 1500, 1800, 2400, 3000, 3500, 4000, 4500, 5000, 6000],
    "Price": [200000, 250000, 280000, 350000, 400000, 450000, 500000, 550000, 600000]
}

df = pd.DataFrame(data)

print("Dataset:")
print(df)

# Features and Target
X = df[['Area']]
y = df['Price']

# Split dataset
X_train, X_test, y_train, y_test = train_test_split(
    X, y, test_size=0.2, random_state=42
)

# Create Linear Regression model
model = LinearRegression()

# Train model
model.fit(X_train, y_train)

# Predict
y_pred = model.predict(X_test)

# Accuracy Metrics
mse = mean_squared_error(y_test, y_pred)
r2 = r2_score(y_test, y_pred)

print("\nModel Coefficient:", model.coef_[0])
print("Model Intercept:", model.intercept_)
print("Mean Squared Error:", mse)
print("R2 Score:", r2)

# Predict new value
new_area = np.array([[3200]])
predicted_price = model.predict(new_area)

print("\nPredicted Price for 3200 sq ft house:", predicted_price[0])
```

Dataset:

	Area	Price
0	1000	200000
1	1500	250000
2	1800	280000
3	2400	350000
4	3000	400000
5	3500	450000
6	4000	500000
7	4500	550000
8	5000	600000
9	6000	700000

Model Coefficient: 99.50551003108215

Model Intercept: 102869.45464820595

Mean Squared Error: 2342401.9569293647

R2 Score: 0.999923513405488

Predicted Price for 3200 sq ft house: 421287.08674766886

C:\Users\MGM\anaconda3\Lib\site-packages\sklearn\base.py:493: UserWarning: X does not have valid feature names, but LinearRegression was fitted with feature names
warnings.warn(

In []: